Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/53**

Paper 5 Probability & Statistics 1 **October/November 2023**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Due to a series-specific issue during the live exam series, all candidates were awarded full marks for questions 1 and 4. This published mark scheme for these questions was created alongside the question paper, but has not been used by examiners.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

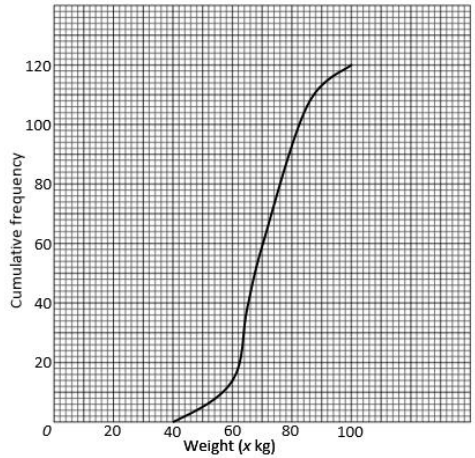
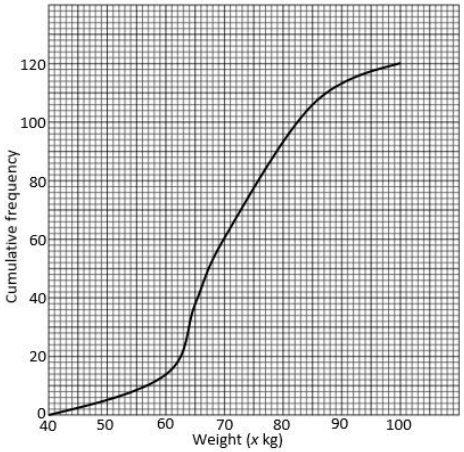
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Question	Answer	Marks	Guidance
1(a)	$2k+6k+12k+20k=1,\ \left[k=\frac{1}{40}\right]$	**M1**	Using sum of probabilities = 1 to form an equation in k. Accept $1\times2\times k + 2\times3\times k + 3\times4\times k + 4\times5\times k = 1$.
	X: 1, 2, 3, 4 P(X): $\frac{2}{40}$, $\frac{6}{40}$, $\frac{12}{40}$, $\frac{20}{40}$ 0.05, 0.15, 0.3, 0.5	**M1**	X: 1, 2, 3, 4 P(X): $2k$, $6k$, $12k$, $20k$ Two correctly linked, accurate probabilities. May be in terms of k. May not be in a table.
		A1	Table with correct X values and correct probabilities.
		3	
1(b)	$[\text{E}(X)=]\ [\text{E}(X)=\frac{1\times2+2\times6+3\times12+4\times20}{40}]\ \frac{2+12+36+80}{40}$	**M1**	$[\text{E}(X)=1\times2k+2\times6k+3\times12k+4\times20k=130k]$ Accept unsimplified expression. May be calculated in variance. FT *their* table with 3 or more probabilities summing to 1 ($0<p<1$). If there are outcomes in the table without probabilities, condone and treat as $p=0$.
	$\left[\text{Var}(X)=\frac{1^2\times2+2^2\times6+3^2\times12+4^2\times20}{40}-(their\ \text{E}(X))^2=\right]$ $\frac{1\times2+4\times6+9\times12+16\times20}{40}-\left(their\frac{13}{4}\right)^2$ $\left[\frac{2+24+108+320}{40}-\left(their\frac{13}{4}\right)^2\right]$	**M1**	$[\text{Var}(X)=\ 1^2\times2k+2^2\times6k+3^2\times12k+4^2\times20k-(130k)^2]$ Appropriate variance formula using *their* $(\text{E}(X))^2$ value. FT *their* table with 3 or more probabilities ($0<p<1$) which need not sum to 1 and the highlighted calculation (or less simplified) seen, Note: if table is correct, $\frac{454}{40}\left(\text{or}\ \frac{227}{20}\ \text{or any calculation}\right)-(their\ \text{E}(X))^2$ implies M1.
	$\text{E}(X)=\frac{13}{4},3\frac{1}{4},3.05\ \ \text{Var}(X)=\frac{63}{80},0.7875$	**A1**	Answers for E(X) and Var(X) must be identified. E(X) may be identified by correct use in variance. Condone E, V, μ, σ etc. If A0 earned, **SC B1** for identified correct final solutions.
		3	

Question	Answer	Marks	Guidance
2(a)	[P(1.42 < X <1.52) =] P($\frac{1.42-1.5}{0.05} < Z < \frac{1.52-1.5}{0.05}$)	**M1**	Use of ± standardisation formula once with 1.5, 0.05 and either 1.42 or 1.52, allow σ^2 or $\sqrt{\sigma}$, no continuity correction.
	[= P($-1.6 < Z < 0.4$) = $\Phi(0.4) + \Phi(1.6) - 1$] = 0.6554 + 0.9452 – 1 or 0.6554 – 0.0548	**M1**	Calculating the appropriate probability area (leading to their final answer, expect > 0.5).
	= 0.601	**A1**	0.6005 < $p \leqslant$ 0.601 **SC B1** for 0.601 with no standardisation seen.
		3	
2(b)	$\left[P(X<0.9) = P\left(Z < \frac{0.9-0.75}{\sigma}\right) = 0.68\right]$ $\frac{0.9-0.75}{\sigma} = 0.468$	**B1**	$0.467 < z \leqslant 0.468$ or $-0.468 \leqslant z < -0.467$ seen
		M1	±standardisation formula with 0.9, 0.75, σ equating to a z-value (not 0.32, 0.68, 0.532, 0.7517, 0.2483, 0.6255,) . Condone continuity correct ±0.05, not $\sigma^2, \sqrt{\sigma}$. Condone $\pm\frac{0.15}{\sigma} = 0.468$.
	$\sigma = 0.321$, $\frac{25}{78}$	**A1**	0.3205 $\leqslant \sigma <$ 0.3215 **SC B1** if M0 www.
		3	

Question	Answer	Marks	Guidance
3(a)	[P(WW) = P(AWW) + P(BWW) =] $\frac{2}{6}\times\frac{8}{15}\times\frac{7}{14}+\frac{4}{6}\times\frac{6}{15}\times\frac{5}{14}$	**M1**	Either $\frac{2}{6}\times\frac{8}{15}\times\frac{7}{14}$ or $\frac{4}{6}\times\frac{6}{15}\times\frac{5}{14}$ seen, accept unsimplified.
		M1	$\frac{q}{6}\times\frac{r}{15}\times\frac{r-1}{14}+\frac{6-q}{6}\times\frac{s}{15}\times\frac{s-1}{14}$ seen, no additional terms, accept unsimplified. Condone $\frac{q}{6}\times\frac{r}{15}\times\frac{r}{15}+\frac{6-q}{6}\times\frac{s}{15}\times\frac{s}{15}$, $1 \leqslant q \leqslant 5$, $1 < r, s < 9$.
	$\left[=\frac{56}{630}+\frac{60}{630}=\frac{4}{45}+\frac{2}{21}\right]=\frac{58}{315}$ or 0.184	**A1**	**SC B1** for 58/315 if either M mark withheld.
		3	

Question	Answer	Marks	Guidance
3(b)	$\left[\text{P(B }\vert\text{WR or RW)} = \frac{\text{P(W \& R from bag B)}}{\text{P(W and R)}} = \right]$ $\frac{\frac{4}{6}\times\frac{6}{15}\times\frac{7}{14}+\frac{4}{6}\times\frac{7}{15}\times\frac{6}{14}}{\frac{2}{6}\times\frac{8}{15}\times\frac{4}{14}+\frac{2}{6}\times\frac{4}{15}\times\frac{8}{14}+\frac{4}{6}\times\frac{6}{15}\times\frac{7}{14}+\frac{4}{6}\times\frac{7}{15}\times\frac{6}{14}}$ or $\frac{2\times\frac{4}{6}\times\frac{6}{15}\times\frac{7}{14}}{2\times\frac{2}{6}\times\frac{8}{15}\times\frac{4}{14}+2\times\frac{4}{6}\times\frac{6}{15}\times\frac{7}{14}}$	**B1**	P(W & R from bag B) $= \frac{2}{3}\times\frac{6}{15}\times\frac{7}{14}+\frac{2}{3}\times\frac{7}{15}\times\frac{6}{14}$ or $2\times\frac{2}{3}\times\frac{6}{15}\times\frac{7}{14}$ $[=\frac{4}{15}\ or\ 0.267]$ Seen alone or as numerator/denominator of conditional probability.
		M1	P(WR or RW) = P(W & R from bag A) + P(W & R from bag B) $= a\times\frac{2}{6}\times\frac{8}{15}\times\frac{4}{14}+a\times\frac{4}{6}\times\frac{6}{15}\times\frac{7}{14}$ or $= a\times\frac{2}{6}\times\frac{8}{15}\times\frac{4}{14}+\textit{their}\ \text{P(W \& R from bag B)}$. a = 1 or 2. [expect $\frac{116}{315}$ or 0.368] Seen alone or as numerator/denominator of conditional probability.
	$\frac{\frac{168}{630}}{\frac{232}{630}} = \frac{\frac{4}{15}}{\frac{116}{315}}$	**M1**	$\frac{\textit{their}\ \text{identified P}(W\ \&R\ from bag\ B)}{\textit{their}\ \text{identified P(WR or RW)}}$ Accept unsimplified.
	$= \frac{168}{232}, \frac{21}{29}$ or 0.724	**A1**	0.7241379 to at least 3SF.
		4	

Question	Answer	Marks	Guidance
4(a)		**M1**	At least 3 points plotted accurately at class upper end points (40,0) (60,14) (65,38) (70,60) (85,106) (100,120). Linear cumulative frequency scale $0 \leqslant cf \leqslant 120$ and linear weight scale $40 \leqslant$ weight(kg) $\leqslant 100$ with at least 3 values identified on each axis. Condone scale reversed.
		**A1**	All points plotted correctly, curve drawn (within tolerance) and joined to (40,0). Axes labelled cumulative frequency (cf), weight (w) and kg (kilograms) – or a suitable title.
		2	
4(b)	[120 × 0.65 =] 78 seen	**M1**	May be implied by use on graph.
	76 [kg]	**A1**	75 < hours < 79. Indication of use of graph required.
		2	

Question	Answer	Marks	Guidance
4(c)	Frequencies: [0] 14 24 22 46 14	**B1**	At least 5 correct frequencies seen (condone omission of 0).
	Midpoints: 20 50 62.5 67.5 77.5 92.5	**B1**	At least 5 correct midpoints seen (condone omission of 20).
	Mean = $\frac{0\times 20+14\times 50+24\times 62.5+22\times 67.5+46\times 77.5+14\times 92.5}{120}$ $=\frac{[0]+700+1500+1485+3565+12950}{120}\left[=\frac{8545}{120}\right]$	**M1**	Correct formula for mean using *their* midpoints and *their* frequencies, implied by $\frac{8545}{120}$ if correct midpoints & frequencies seen. May be gained in variance calculation. If midpoints not clearly identified, condone midpoints ± 0.5.
	= 71.2	**A1**	Accept $\frac{1709}{24}$, $71\frac{5}{24}$ or 71.208333 to at least 3SF. If M0 scored, **SC B1** for $\frac{1709}{24}$, $71\frac{5}{24}$ or 71.208333 to at least 3SF www.
	Variance = $\frac{0\times 20^2+14\times 50^2+24\times 62.5^2+22\times 67.5^2+46\times 77.5^2+14\times 92.5^2}{120}-71.2^2$ $\frac{[0]+35000+93750+100237.5+276287.5+119787.5}{120}-\left(\frac{8545}{120}\right)^2$ $[=138.23]$	**M1**	Correct formula for variance using *their* midpoints, *their* frequencies and *their* mean. Implied by $\frac{625062.5}{120}-\left(\frac{8545}{120}\right)^2$ if correct midpoints & frequencies seen.
	Standard deviation = 11.8	**A1**	11.757016 to at least 3SF.
		6	

Question	Answer	Marks	Guidance
5(a)(i)	**Method 1**		
	$[\mathrm{P}(2 \leqslant X \leqslant 6) = \mathrm{P}(X \leqslant 6) - \mathrm{P}(X \leqslant 1) =] 1 - (0.7)^6 - (1 - 0.7)$	**M1**	$1 - 0.7^n$ seen, $n = 5, 6$.
	= 0.582	**A1**	www 0.582351 to at least 3SF.
	Method 2		
	P(X= 2, 3, 4, 5, 6) $= 0.7 \times 0.3 + 0.7^2 \times 0.3 + 0.7^3 \times 0.3 + 0.7^4 \times 0.3 + 0.7^5 \times 0.3$ = 0.21 + 0.147 + 0.1029 + 0.07203 + 0.050421	**M1**	Sum of first 4 or 5 correct terms – no incorrect terms.
	= 0.582	**A1**	www 0.582351 to at least 3SF.
		2	
5(a)(ii)	$3\frac{1}{3}$	**B1**	Condone 3.33, $3.\dot{3}$ or $\frac{10}{3}$ – NOT $\frac{1}{0.3}$.
		1	

Question	Answer	Marks	Guidance
5(b)	**Method 1**		
	[P(3, 4, 5) =] ${}^5C_3(0.3)^3(0.7)^2 + {}^5C_4(0.3)^4(0.7)^1 + {}^5C_5(0.3)^5(0.7)^0$	**M1**	One term seen ${}^5C_x\ (p)^x(1-p)^{5-x},\ 0<p<1,\ x\neq 0,5.$
	= 0.1323 + 0.02835 + 0.00243	**A1**	Correct expression, accept unsimplified, no terms omitted leading to final answer.
	= 0.163, $\frac{4077}{25000}$	**B1**	0.16308 to at least 3SF.
	Method 2		
	[1 – P(0, 1, 2) = $1-({}^5C_0(0.3)^0(0.7)^5 + {}^5C_1(0.3)^1(0.7)^4 + {}^5C_2(0.3)^2(0.7)^3)$	**M1**	One term ${}^5C_x\ (p)^x(1-p)^{5-x},\ 0<p<1,\ x\neq 0,5.$
	= 1 – (0.16807 + 0.36015 + 0.3087)	**A1**	Correct expression, accept unsimplified, no terms omitted leading to final answer.
	= 0.163, $\frac{4077}{25000}$	**B1**	0.16308 to at least 3SF.
		3	

Question	Answer	Marks	Guidance
5(c)	[Mean = $75\times0.3=$] 22.5 [Var = $75\times0.3\times0.7=$] 15.75	**B1**	22.5, 22½ and 15.75, $15\frac{3}{4}$ seen, allow unsimplified. ($\sigma = \frac{3\sqrt{7}}{2}$ or 3.9686269… to at least 3SF implies correct variance)
	$[\mathrm{P}(X>20)=]\ \mathrm{P}\left(Z>\frac{20.5-22.5}{\sqrt{15.75}}\right)$	**M1**	Substituting their μ and σ into ±standardisation formula (any number for 20.5), not σ^2 not $\sqrt{\sigma}$.
		M1	Using continuity correction 19.5 or 20.5 in *their* standardisation formula.
	$[\mathrm{P}(Z>-0.504)=\Phi(0.504)]$ = 0.693	**M1**	Appropriate area Φ, from final process, must be a probability. Expect final answer > 0.5. Note: correct final answer implies this M1.
		A1	$0.6925 < p \leqslant 0.693$
		5	

Question	Answer	Marks	Guidance
6(a)(i)	**Method 1**		
	$6! \times 2^6$	**M1**	$6! \times a$, a integer > 1.
		M1	$b \times 2^6$, b integer $\geqslant 1$.
	= 46080	**A1**	Accurate answer required. **SC B1** for 46080 if M0 M0 www.
	Alternative method for question 6(a)(i)		
	$12 \times 10 \times 8 \times 6 \times 4 \times 2$	**M1**	$c \times d \times e \times f \times g \times h$ $2 \leqslant c,d,e,f,g,h$ (different integers) $\leqslant 12$
		M1	Correct unsimplified.
	= 46080	**A1**	Accurate answer required. **SC B1** for 46080 if M0 M0 www.
		3	
6(a)(ii)	$5! \times 5! \times 2 \times 2$	**M1**	$5! \times 5! \times k$, k positive integer, 1 may be implied (no adding/subtracting).
	= 57600	**A1**	
		2	

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="5">6(b)</td><td colspan="3">Method 1 probabilities of J & K being placed:</td></tr>
<tr><td rowspan="3"><table><tr><td>In the group of 5</td><td>$\frac{5}{12}\times\frac{4}{11}$</td><td>$\left[=\frac{20}{132},\frac{5}{33}\right]$</td></tr><tr><td>In the group of 4</td><td>$\frac{4}{12}\times\frac{3}{11}$</td><td>$\left[=\frac{12}{132},\frac{1}{11}\right]$</td></tr><tr><td>In the group of 3</td><td>$\frac{3}{12}\times\frac{2}{11}$</td><td>$\left[=\frac{6}{132},\frac{1}{22}\right]$</td></tr></table></td><td>B1</td><td>Correct probability for one identified scenario.</td></tr>
<tr><td>M1</td><td>Denominator 12×11 for all probabilities, (1, 2 or 3 scenarios).</td></tr>
<tr><td>A1</td><td>3 correct probabilities, accept unsimplified.</td></tr>
<tr><td>$\frac{5}{12}\times\frac{4}{11}+\frac{4}{12}\times\frac{3}{11}+\frac{3}{12}\times\frac{2}{11}$</td><td>M1</td><td>Adding probabilities for 3 correct scenarios.</td></tr>
<tr><td>$\frac{19}{66}$, 0.288</td><td>A1</td><td>0.2878787 to at least 3SF.</td></tr>
</table>

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="6">6(b)</td><td colspan="3">Method 2 number of arrangements of J & K being placed:</td></tr>
<tr><td rowspan="3"><table><tr><td>In the group of 5</td><td>$^{10}C_3 \times {}^7C_4$</td><td>[= 120 × 35 = 4200]</td></tr><tr><td>In the group of 4</td><td>$^{10}C_2 \times {}^8C_5$</td><td>[= 45 × 56 = 2520]</td></tr><tr><td>In the group of 3</td><td>$^{10}C_1 \times {}^9C_5$</td><td>[= 10 × 126 = 1260]</td></tr></table></td><td>B1</td><td>Correct value of one identified scenario seen, accept unsimplified.</td></tr>
<tr><td>M1</td><td>$^{12}C_a \times {}^{12-a}C_b$, a = 3, 4, 5; b = 3, 4, 5 ($a \neq b$)</td></tr>
<tr><td>[Total number of ways of arranging the 3 groups =]
$^{12}C_5 \times {}^7C_4 = 792 \times 35 = 27720$
or $^{12}C_3 \times {}^9C_4$ or $^{12}C_4 \times {}^8C_5$</td><td>A1</td><td>27720 Seen alone or as denominator of probability –accept unsimplified.
SC B1 if M0.</td></tr>
<tr><td>4200 + 2520 + 1260 = 7980</td><td>M1</td><td>Values of 3 correct scenarios added, accept unsimplified – or correct.</td></tr>
<tr><td>[Probability =] $\frac{7980}{27720}$, $\frac{19}{66}$, 0.288</td><td>A1</td><td>0.2878787 to at least 3SF.</td></tr>
<tr><td></td><td>5</td><td>Note, alternative arrangement calculations possible e.g.<table><tr><td>In the group of 5</td><td>$^{10}C_3 \times {}^7C_4$</td><td>[= 120 × 35 = 4200]</td></tr><tr><td>In the group of 4</td><td>$^{10}C_5 \times {}^5C_2$</td><td>[= 252× 10 = 2520]</td></tr><tr><td>In the group of 3</td><td>$^{10}C_5 \times {}^5C_4$</td><td>[= 252 × 5 = 1260]</td></tr></table></td></tr>
</table>

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